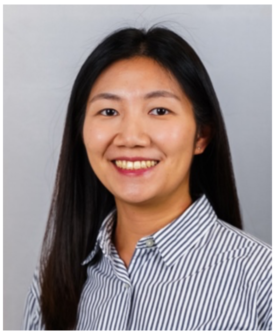

# Ming-Hsuan Yen


**PhD in Geophysics | Natural Hazards | Geospatial data Analytics**

Potsdam, Germany | minghsuanyen@gmail.com | 0152 2269 4099 |
website: https://mingslab.com/ | No visa sponsorship required


## Professional Summary

PhD-trained scientist with 8+ years of experience turning large, complex geospatial and time-series datasets into actionable risk insights. Skilled in Python-based data pipelines, statistical and numerical modeling, and validating model outputs against real-world data. Proven track record of translating technical findings for non-specialist audiences, including government agencies and industry stakeholders. Experienced in cross-institutional, international collaboration across Europe, Asia, and the Pacific.

## Core Competencies

- AI-assisted workflow/ Agentic AI
- Data pipeline development & automation
- Statistical & numerical modeling
- Time-series data analysis
- Risk quantification & uncertainty analysis
- Python / scientific computing
- Geospatial data analysis (GIS)
- Cross-functional & international collaboration
- Technical reporting & stakeholder communication
- Scientific & data quality control

## Professional Experience

### Scientific Researcher

***Seismic Hazard and Risk Dynamics, GFZ-Potsdam, Potsdam, Germany (2019 – 2026)***

**Projects: New challenges for Urban Engineering Seismology (H2020-MSCA-ITN-2018)*

- Built reproducible Python pipelines to integrate large, heterogeneous datasets across formats and coordinate systems, enabling consistent analysis at a European scale.
- Developed and validated statistical and physics-based models against empirical data from multiple international networks (Taiwan, Japan, New Zealand, Europe), ensuring model reliability and accuracy.
- Conducted numerical simulations to support risk assessment, translating complex outputs into actionable insights for hazard evaluation.
- Communicated technical findings to non-specialist and policy audiences through reports, presentations, and international conferences.
- Published 5 peer-reviewed research articles (selected list below).

### Research Intern

***Aon, Prague, Czechia (Apr 2022 – May 2022)***

- Curated and quality-controlled multi-country datasets to support risk modeling, contributing to underwriting-relevant analysis and discussions.

### Research Assistant

***Earthquake Disaster & Risk Evaluation and Management Center, National Central University, Taiwan (2016 – 2019)***

**Projects:*

1. *Understanding Earthquake Hazard for Earthquake System Science*
2. *Taiwan Earthquake Model: from Earthquake Hazard and Risk Scenario*
3. *Host Rock Characterization and Evaluation, Spent Nuclear Fuel Final Disposal Program (2015–2018)*

- Conducted numerical simulations to support regional hazard model development and international collaborative research (Taiwan, Japan, New Zealand).

- Performed advanced simulations to determine key parameters for a high-stakes site risk evaluation, validated against empirical data.
- Collaborated with a multidisciplinary team (geology, seismology, engineering) to integrate cross-domain evidence into risk assessments, delivering findings to government agencies.

## Education

**Dr. rer. nat., Geophysics**
***Universität Potsdam, Germany (2019 – 2024)***
*EU Marie Skłodowska-Curie Fellow (URBASIS). Graduated magna cum laude.
*Thesis: Earthquakes Source Parameters and Near-source Velocity Pulses Variabilities in the Context of Ground Motion Prediction

**M.Sc., Geophysics**
***National Central University, Taiwan (2014 – 2016)***
*Thesis: The source rupture analysis and 3D seismic wave simulations of the 1935 Hsinchu-Taichung Earthquake

**B.Sc., Earth Science**
***National Central University, Taiwan (2010 – 2014)***

## Technical Skills

| | |
|---|---|
| **Programming** | Python, FORTRAN, MATLAB, SQL, TypeScript |
| **Tools** | QGIS, Jupyter, Linux/Unix, Git/GitHub, agentic AI |
| **Languages** | English (advanced), German (intermediate), Mandarin (native) |

## Selected Publications **(Full publication list available upon request.)**

Yen et al. (2025) - An analysis of directivity pulses using empirical data and dynamic rupture simulations of the 2023 Kahramanmaraş earthquake doublet. *Earthquake Spectra. doi: https://doi.org/10.1177/87552930241305012*

Yen et al. (2024) - Source parameters and scaling relationships of stress drop for shallow crustal earthquakes in Western Europe. *Journal of Seismology. doi: https://doi.org/10.1007/s10950-023-10188-y*

Yen et al. (2022) - Within- and between-event variabilities of strong-velocity pulses of moderate earthquakes within dense seismic arrays. *Bulletin of the Seismological Society of America. doi: https://doi.org/10.1785/0120200376*

## Certificates

Natural Disaster and Climate Change Risk Assessment (2026)
AI Software Development (2026)
IBM Data Science Professional Certificate (2025)